

Insurance Claim Prediction

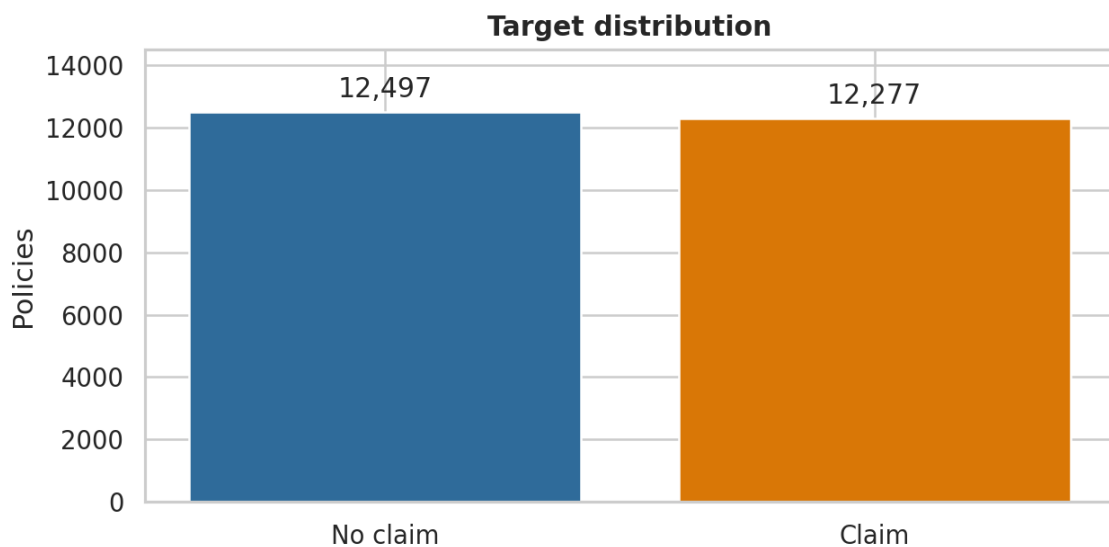
Interpretable classification for automobile insurance risk

Portfolio report

LDATS2350 Data Mining

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Objective: predict whether an automobile insurance policy will generate a claim during its coverage period, while preserving enough transparency to explain the main risk associations.



Dataset: 24,774 policies, 10 predictors, and one binary target. After removing 21 exact duplicates, 24,753 observations remain.

1. Business context

Insurers need reliable estimates of claim risk to support prevention, portfolio monitoring, and pricing decisions. This project compares five classification methods using policyholder, vehicle, coverage, and geographic characteristics.

Target and predictors

The target **claimNumbMD** equals 1 when at least one claim occurred and 0 otherwise. Predictors include underwriting year, gender, vehicle type and category, occupation, age, years of driving experience, vehicle value, coverage, and population density.

Data-quality audit

Check	Result	Treatment
Rows	24,774	-
Exact duplicates	21	Removed before modelling
Missing values	0	Pipeline supports imputation
Class balance	50.5% no claim / 49.5% claim	Stratified train-test split

Exploratory findings

The original exploratory analysis found limited correlation among continuous predictors. Vehicle value was right-skewed and therefore receives a square-root transformation. Chi-square tests indicated associations between claim occurrence and all tested categorical variables; these tests indicate association, not causality or operational usefulness.

Observed patterns suggested lower predicted risk for older and retired policyholders, and higher predicted risk for male drivers, unemployed policyholders, type-E vehicles, and denser areas. These results require fairness and legal review before real insurance use.

2. Modelling approach

Data are split into 70% training and 30% testing sets with class stratification. Every learned preprocessing step is contained in a scikit-learn pipeline: median imputation and standardisation for numeric variables, square-root transformation for vehicle value, and one-hot encoding for categorical variables.

This corrects a weakness in the original coursework, where scaling occurred before the train-test split. Learning preprocessing only from the training sample prevents information leakage into the held-out evaluation.

Models compared

Logistic regression provides a transparent linear baseline. A decision tree captures interactions and non-linearity. K-nearest neighbours offers a non-parametric comparison. A multilayer perceptron models complex relationships. Gaussian Naive Bayes provides a fast probabilistic benchmark. Tuned models use five-fold cross-validation on the training set, with ROC AUC as the selection metric.

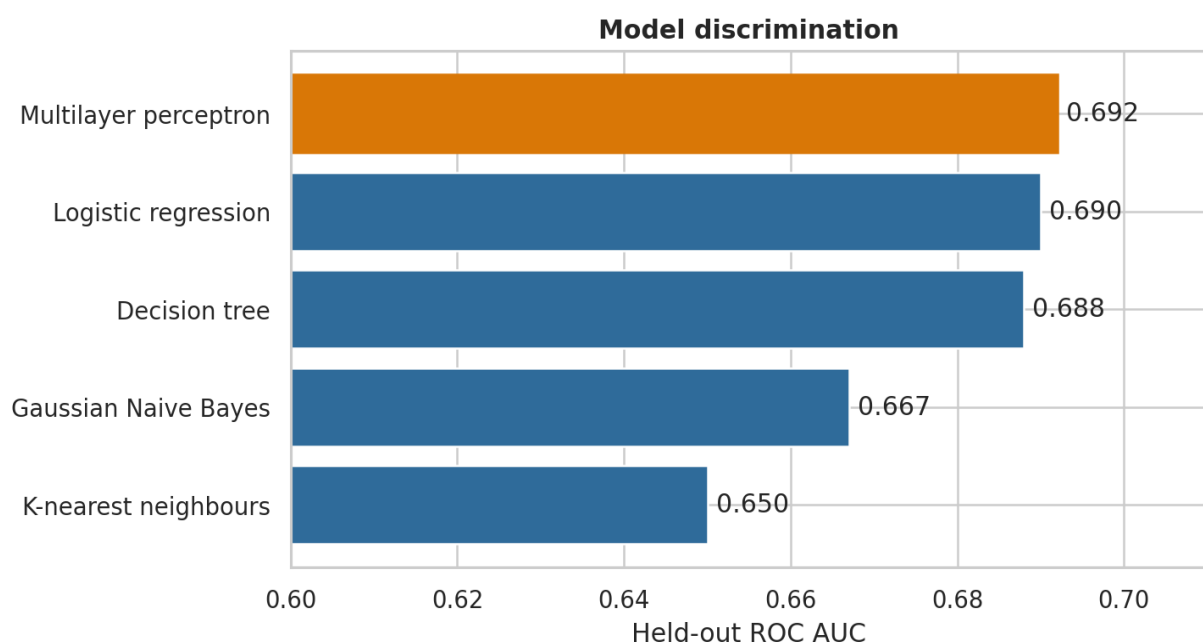


Figure 1. Held-out ROC AUC; higher values indicate better discrimination.

3. Results

The MLP ranks first by ROC AUC, but its advantage over logistic regression is only 0.002. Logistic regression provides virtually the same discrimination with much clearer interpretation. Naive Bayes achieves the highest recall, showing the cost of prioritising claim detection: more false positives and lower precision.

Model	Accuracy	Precision	Recall	F1	ROC AUC
Multilayer perceptron	0.639	0.639	0.621	0.630	0.692
Logistic regression	0.637	0.627	0.660	0.643	0.690
Decision tree	0.640	0.645	0.606	0.625	0.688
Gaussian Naive Bayes	0.608	0.576	0.790	0.666	0.667
K-nearest neighbours	0.609	0.607	0.596	0.601	0.650

Recommended model

Logistic regression is the recommended portfolio model. It reaches ROC AUC 0.690, accuracy 0.637, recall 0.660, and F1 0.643. The MLP's marginal AUC advantage does not outweigh the loss of transparency for this educational insurance-risk application.

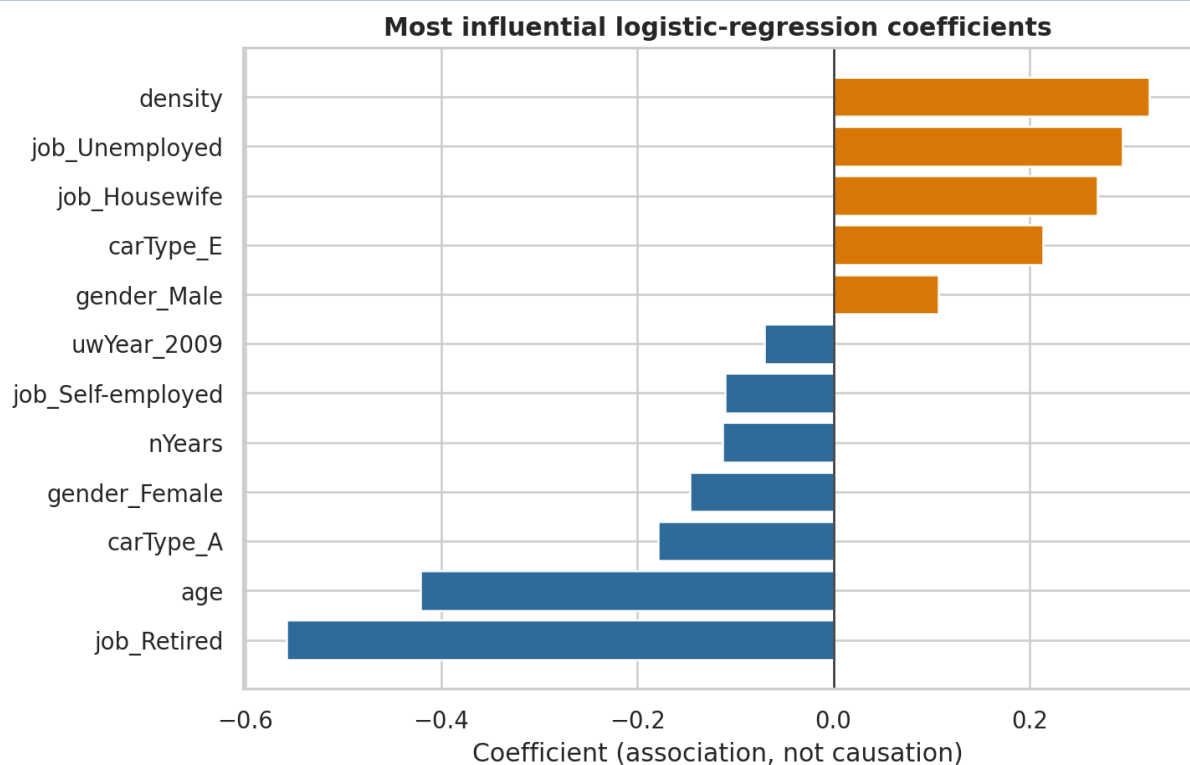


Figure 2. Largest logistic-regression coefficients by absolute magnitude.

4. Conclusions and next steps

The project demonstrates a complete classification workflow, from data auditing and exploration to leakage-safe modelling, comparison, and interpretation. The available variables contain useful signal, but an AUC near 0.69 remains moderate and is insufficient for unsupervised production decisions.

What worked

- A reproducible pipeline eliminated preprocessing leakage.
- A common held-out set made model comparison consistent.
- Logistic regression preserved interpretability with near-best discrimination.
- Recall and precision exposed operational trade-offs hidden by accuracy alone.

Recommended extensions

- Add prior claims, mileage, exposure duration, driving behaviour, and richer vehicle information.
- Use repeated or nested cross-validation to quantify performance uncertainty.
- Calibrate predicted probabilities and select thresholds using claim costs.
- Evaluate subgroup error rates and fairness before considering deployment.
- Monitor drift and recalibrate any model used over time.

Responsible-use note

This is an educational project, not a production underwriting system. Gender and occupation can raise discrimination, fairness, and regulatory concerns. Real-world use would require legal review, documented governance, human oversight, calibration, bias testing, monitoring, and secure data management.

Reproducibility

The complete source analysis is available in **analysis.qmd**, with dependencies in **requirements.txt** and data in **data/dataSetJune2025.csv**. Random seeds and the stratified split are fixed for repeatable results.